

Constructing Models from Valuations

Miniseminar on models of curves

Max Schwegele

University Ulm

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In this talk, we want to finish the proof of our main theorem. The notation will be the same as in the previous talk. We fix a "nice" curve X — smooth, projective, and geometrically irreducible — defined over a field K that satisfies our assumptions (e.g., a finite extension of $\mathbb{Q}_p$). We study the models $\mathcal{X}$ of X , which are integral, proper, flat, and normal $\mathcal{O}_K$ -schemes, with an identification of the generic fiber with X .

Theorem 1 *The map*

$$\mathcal{X} \mapsto V(\mathcal{X})$$

defines a monotone bijection between

- *the set of isomorphism classes of models of X , and*
- *the set of finite, nonempty subsets of $V_{\text{II}}(F_X)$.*

In the last talk, we have seen that the map $[\mathcal{X}] \mapsto V(\mathcal{X})$ is monotone and injective. It remains to show the surjectivity of the map, i.e., for a given finite, nonempty set of valuations $V \subset V_{\text{II}}(F_X)$, there is a model $\mathcal{X}$ of X such that $V = V(\mathcal{X})$. We will prove that in two steps, which we state in the following proposition.

Proposition 2 ([1, 3.14, 3.16], [2, 8.3.36])

- (i) *Given model $\mathcal{X}$ and a valuation $v \in V_{\text{II}}(F_X)$, there exists a modification $\varphi: \mathcal{X}' \rightarrow \mathcal{X}$ such that $v \in V(\mathcal{X}')$.*
- (ii) *Given a model $\mathcal{X}$ and a nonempty subset $V \subset V(\mathcal{X})$, there exists a modification $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$ such that $V = V(\mathcal{X}')$.*

After proving Proposition 2, the proof is complete: Starting with a model $\mathcal{X}$ of X , we can apply (i) for every $v \in V \subset V_{\text{II}}(F_X)$ to obtain a new model that induces all valuations in V , but possibly more. We then apply (ii) to construct a model that induces exactly the valuations in V .

Proof of (i)

We fix a model $\mathcal{X}$ of X and a type II valuation $v \in V_{\text{II}}(F_X)$.

Definition 3 ([2, 8.3.17]) A point $x \in \mathcal{X}$ is called a center of v if there exists a morphism $f: \text{Spec } \mathcal{O}_v \rightarrow \mathcal{X}$ of $\mathcal{O}_K$ -schemes, which maps the closed point to x and makes the following diagram commute:

$$\begin{array}{ccc} \text{Spec } \mathcal{O}_v & \xrightarrow{f} & \mathcal{X} \\ & \nwarrow \quad \nearrow & \\ & \text{Spec } F_X & \end{array}$$

Here, the left map is induced by the inclusion $\mathcal{O}_v \hookrightarrow \text{Frac } \mathcal{O}_v = F_X$, and the right map comes from the identification of X with the special fiber $\mathcal{X}_s$. Specifically, it is the canonical map $\text{Spec } F_X \simeq \text{Spec } \mathcal{O}_{\mathcal{X}, \zeta} \rightarrow \mathcal{X}$, where ζ is the generic point of $\mathcal{X}$.

Lemma 4 *There exists a unique center $x \in \mathcal{X}$ of v .*

Proof. By the valuative criterion for properness, such a center exists, and the morphism f is unique (see also [3, Section 01J5, Lemma 01J6]):

$$\begin{array}{ccc} \text{Spec } F_X & \xrightarrow{\quad} & \mathcal{X} \\ \downarrow & \searrow f & \downarrow \\ \text{Spec } \mathcal{O}_v & \xrightarrow{\quad} & \text{Spec } \mathcal{O}_K. \end{array}$$

□

Lemma 5 ([1, 3.11]) *We have $v \in V(\mathcal{X})$ if and only if the center x of v is of codimension one. In that case x is the generic point on the special fiber which induces v .*

Proof. Assume $v \in V(\mathcal{X})$. Then v is induced by a generic point $\eta \in \mathcal{X}_s$ on the special fiber. The canonical morphism $\text{Spec } \mathcal{O}_{\mathcal{X}, \eta} \rightarrow \mathcal{X}$ shows that v has center η , which is indeed a point of codimension one. Conversely, suppose $v \in V_{\Pi}(F_X)$ has center $x \in \mathcal{X}$ of codimension one. The morphism $\text{Spec } \mathcal{O}_v \rightarrow \mathcal{X}$ uniquely factors through $\text{Spec } \mathcal{O}_v \rightarrow \text{Spec } \mathcal{O}_{\mathcal{X}, x}$, so $\mathcal{O}_v$ dominates $\mathcal{O}_{\mathcal{X}, x}$. Since these are both valuation rings with common fraction field K , it follows that these are in fact equal. Thus, $v \in V(\mathcal{X})$. □

Lemma 6 ([1, 3.12, 3.13]) *Let $x \in \mathcal{X}$ be the center of v , and let $f \in \mathcal{O}_v^\times \subset F_X$ be such that the reduction $\bar{f} \in k(v)$ is transcendental over k . Then:*

- (a) *If $f \in \mathcal{O}_{\mathcal{X}, x}$, then $v \in V(\mathcal{X})$*
- (b) *If the rational map $\mathcal{X} \dashrightarrow \mathbb{P}_{\mathcal{O}_K}^1$ induced by f is defined at the center x of v , then $v \in V(\mathcal{X})$.*

Proof. (a) By Lemma 5, it suffices to show that x is a point of codimension one. Choose an open affine set $\text{Spec } A = U \subset \mathcal{X}$ such that $x \in U$ and $f \in A$. Denote by $\mathfrak{p} \subset A$ the prime ideal corresponding to x . Since v is non-trivial, the center x is not the generic point of $\mathcal{X}$, and so the codimension is not zero. If the dimension of $A/\mathfrak{p}$ is at least one, then the inequality

$$\dim(A_{\mathfrak{p}}) + \dim(A/\mathfrak{p}) \leq \dim(A) = 2$$

implies

$$\text{codim}(\overline{\{x\}}, \mathcal{X}) = \dim(A_{\mathfrak{p}}) \leq 2 - \dim(A/\mathfrak{p}) \leq 2 - 1 = 1,$$

and hence equal to one. Thus, it suffices to show that $A/\mathfrak{p}$ is not field. If $A/\mathfrak{p}$ were a field, it would be a finite extension of k , but that is impossible since $\bar{f} \in A/\mathfrak{p} \subset k(v)$ is transcendental over k .

- (b) It suffices to show, using (a), that $f \in \mathcal{O}_{\mathcal{X}, x}$. Let $P = V(f^{-1})$ and $Z = V(f)$ be the sets of poles and zeros of f , respectively. Then the rational map $\mathcal{X} \dashrightarrow \mathbb{P}_{\mathcal{O}_K}^1$ is obtained by glueing the morphisms

$$\mathcal{X} \setminus P \rightarrow \text{Spec } \mathcal{O}_K[t] \subset \mathbb{P}_{\mathcal{O}_K}^1, \quad \text{and} \quad \mathcal{X} \setminus Z \rightarrow \text{Spec } \mathcal{O}_K[t^{-1}] \subset \mathbb{P}_{\mathcal{O}_K}^1,$$

which are induced by

$$\begin{array}{ccc} \mathcal{O}_K[t] \rightarrow \Gamma(\mathcal{X} \setminus P, \mathcal{O}_{\mathcal{X}}), & \text{and} & \mathcal{O}_K[t^{-1}] \rightarrow \Gamma(\mathcal{X} \setminus Z, \mathcal{O}_{\mathcal{X}}), \\ t \mapsto f & & t^{-1} \mapsto f^{-1} \end{array}$$

and is defined on $U := (\mathcal{X} \setminus P) \cup (\mathcal{X} \setminus Z) = \mathcal{X} \setminus (P \cap Z)$. By assumption, $x \in U$. If $x \in \mathcal{X} \setminus P$, the claim follows immediately, as

$$f \in \Gamma(\mathcal{X} \setminus P, \mathcal{O}_{\mathcal{X}}) \subset \mathcal{O}_{\mathcal{X},x}.$$

If $x \in \mathcal{X} \setminus Z$, then

$$f^{-1} \in \Gamma(\mathcal{X} \setminus Z, \mathcal{O}_{\mathcal{X}}) \subset \mathcal{O}_{\mathcal{X},x}.$$

The image of f^{-1} under the local ring homomorphism $\mathcal{O}_{\mathcal{X},x} \rightarrow \mathcal{O}_v$ is a unit by assumption, hence f^{-1} is also a unit in $\mathcal{O}_{\mathcal{X},x}$. Consequently $f \in \mathcal{O}_{\mathcal{X},x}$. $\square$

We can now finish the proof of Proposition 2 (i).

Proof (of Proposition 2 (i)). Let $f \in \mathcal{O}_v^\times \subset F_X$ be such that the reduction $\bar{f} \in k(v)$ is transcendental over k . We also write $f: \mathcal{X} \dashrightarrow \mathbb{P}_{\mathcal{O}_K}^1$ for the rational map induced by f , and let U be its domain of definition. Define $\Gamma_f \subset \mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}_{\mathcal{O}_K}^1$ to be the graph of f . Recall that it is defined to be the closure of the graph $f|_U$ in $\mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}_{\mathcal{O}_K}^1 \supset U \times_{\mathcal{O}_K} \mathbb{P}_{\mathcal{O}_K}^1$. This is almost a model of X , as it satisfies all the properties of a model except that it might not be normal.

$$\begin{array}{ccccc} & & \mathcal{X}' & & \\ & & \downarrow & & \\ \mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}_{\mathcal{O}_K}^1 & \supseteq & \Gamma_f & \longrightarrow & \mathbb{P}_{\mathcal{O}_K}^1 \\ & & \downarrow & \nearrow f & \\ & & \mathcal{X} & & \end{array}$$

The first projection $\mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}_{\mathcal{O}_K}^1 \rightarrow \mathcal{X}$ induces a birational morphism $\Gamma_f \rightarrow \mathcal{X}$. The second projection induces a morphism $\Gamma_f \rightarrow \mathbb{P}_{\mathcal{O}_K}^1$ which extends $f: \mathcal{X} \dashrightarrow \mathbb{P}_{\mathcal{O}_K}^1$. Let $\mathcal{X}' \rightarrow \Gamma_f$ denote the normalization of Γ_f . This gives a model of $\mathcal{X}'$ for which $f: \mathcal{X}' \rightarrow \mathbb{P}_{\mathcal{O}_K}^1$ is a morphism. By Lemma 6 (b), this implies that $v \in V(\mathcal{X}')$. $\square$

Proof of (ii)

We fix a model $\mathcal{X}$ of X and a nonempty subset $V \subset V(\mathcal{X})$. Denote by $\mathcal{E}$ the set of irreducible (integral) components of $\mathcal{X}_s$ that induce the valuations in $V(\mathcal{X}) \setminus V$.

Definition 7 ([2, 8.3.27]) A model $\mathcal{X}'$ together with a modification $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$ such that for every integral vertical curve E on $\mathcal{X}$, the set $\varphi(E)$ is a point if and only if $E \in \mathcal{E}$ is called a contraction of the $E \in \mathcal{E}$.

It can be shown that if a contraction exists, it is unique up to unique isomorphism. Our goal now is to show that a contraction of $\mathcal{E}$ exists. Once this is done, the proof is complete, since for the contraction $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$, we have by construction $V(\mathcal{X}') = V$.

Lemma 8 ([2, 8.3.29, 8.3.3]) *Let $\mathcal{L}$ be an invertible sheaf on $\mathcal{X}$ generated by global sections $s_0, \dots, s_n$. Consider the morphism $f: \mathcal{X} \rightarrow \mathbb{P}_{\mathcal{O}_K}^n$ associated to these sections. Let E be a vertical curve of $\mathcal{X}_s$ with the reduced induced closed subscheme structure. Then $f(E)$ is reduced to a point if and only if $\mathcal{L}|_E \simeq \mathcal{O}_E$.*

Proposition 9 ([2, 8.3.30]) *The following are equivalent.*

- (a) *The contraction $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$ of $\mathcal{E}$ exists.*
- (b) *There exists a Cartier divisor D on $\mathcal{X}$ such that $\deg(D|_{\mathcal{X}_s}) > 0$, that $\mathcal{L} := \mathcal{O}_{\mathcal{X}}(D)$ is generated by its global sections, and that for any integral vertical curve E , we have $\mathcal{L}|_E \simeq \mathcal{O}_E$ if and only if $E \in \mathcal{E}$.*

Proof. We only sketch (b) $\implies$ (a). For the other implication and details see [2, 8.3.30].

As $\mathcal{X}_s$ is an integral projective curve over a field, $\mathcal{L}|_{\mathcal{X}_s}$ is an ample divisor ([2, Proposition 7.5.5]). Replacing D by a positive multiple (which does not change the set $\mathcal{E}$), we can suppose that $\mathcal{L}|_{\mathcal{X}_s}$ is very ample.

Let $f: \mathcal{X} \rightarrow \mathbb{P}_{\mathcal{O}_K}^n$ be a morphism associated to sections $s_0, \dots, s_n \in \Gamma(\mathcal{X}, \mathcal{L})$ which generate $\mathcal{L}$. Let Z be the closed subset $f(\mathcal{X}) \subset \mathbb{P}_{\mathcal{O}_K}^n$ endowed with the reduced (hence integral) scheme structure. Then f induces a dominant morphism $\mathcal{X} \rightarrow Z$ that we still denote by f . This is an isomorphism on the generic fiber because $\mathcal{L}|_{\mathcal{X}_s}$ is very ample. In particular, $f: \mathcal{X} \rightarrow Z$ is birational. Denote by $\pi: \mathcal{X}' \rightarrow Z$ the normalization morphism. As $\mathcal{X}$ is normal, f uniquely factors into $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$ followed by $\pi: \mathcal{X}' \rightarrow Z$.

$$\begin{array}{ccc} & & \mathcal{X}' \\ & \nearrow \varphi & \downarrow \pi \\ \mathcal{X} & \xrightarrow{f} & Z \end{array}$$

Let E be an integral vertical curve on $\mathcal{X}$. Then we have

$$f(E) \text{ is a point} \xLeftrightarrow{\text{Lemma 8}} \mathcal{L}|_E \simeq \mathcal{O}_E \xLeftrightarrow{\text{assumption}} E \in \mathcal{E}.$$

We claim that $\pi: \mathcal{X}' \rightarrow Z$ is a finite morphism. This then completes the proof because $f(E)$ is reduced to a point if and only if $\varphi(E)$ is a point. The morphism f is projective ([2, Corollary 3.3.32(e)]), so $\mathcal{A} := f_*\mathcal{O}_{\mathcal{X}}$ is a coherent sheaf on Z ([2, Corollary 5.3.5]). Furthermore, $\mathcal{A}$ is a sheaf of integrally closed algebras because $\mathcal{X}$ is normal. As a result, the normalization morphism $\pi: \mathcal{X}' \rightarrow Z$ coincides with the canonical morphism $\text{Spec } \mathcal{A} \rightarrow Z$ and is a finite morphism ([2, Exercise 5.1.17]). $\square$

For the construction of a suitable Cartier divisor, we need also the following technical lemma.

Lemma 10 ([2, 8.3.32]) *Let D be an effective horizontal Cartier divisor on $\mathcal{X}$, and $\mathcal{L} = \mathcal{O}_{\mathcal{X}}(D)$. Then there exists an $m_0 \geq 1$ such that $\mathcal{L}^{\otimes m}$ is generated by its global sections for every $m \geq m_0$.*

Proposition 9 and Lemma 10 show that for a contraction morphism to exist, it suffices that there exists an effective Cartier divisor that meets the closed fibers in a suitable way. To construct such a divisor, it is important that the ring $\mathcal{O}_K$ is complete, which implies it is Henselian.

Definition 11 ([2, 8.3.33], [3, Lemma 04GG]) Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k := A/\mathfrak{m}$. We say that A is Henselian if one of the following equivalent conditions is fulfilled:

- Hensel's Lemma holds: For every monic $f \in A[T]$, and every simple root $a_0 \in k$ of $\bar{f} \in k[T]$, there exists a lift $a \in A$ (so $\bar{a} = a_0$) such that $f(a) = 0$.
- Every finite A -algebra is a direct sum of local A -algebras.

Lemma 12 ([2, 8.3.35]) Let $x \in \mathcal{X}_s$ be a closed point on the special fiber. Then the following properties are true.

- (a) There exists an irreducible horizontal Weil divisor Δ containing x .
- (b) There exists an effective horizontal Cartier divisor D such that $\mathcal{X}_s \cap \text{Supp } D = \{x\}$.

Proof. (a) Let $\mathfrak{m}_x$ be the maximal ideal of $\mathcal{O}_{\mathcal{X},x}$ and denote by π a uniformizer of $\mathcal{O}_K$. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ be the prime ideals of $\mathcal{O}_{\mathcal{X},x}$ that are minimal among those containing $\pi\mathcal{O}_{\mathcal{X},x}$. As $\dim \mathcal{O}_{\mathcal{X},x} = 2$ ([2, Proposition 8.3.4]), Krull's principal ideal theorem implies that $\mathfrak{m}_x \neq \mathfrak{p}_i$. There therefore exists an

$$f \in \mathfrak{m}_x \setminus (\cup_i \mathfrak{p}_i).$$

Thus f is a regular element and $V(f)$ does not contain any irreducible component of $\text{Spec } \mathcal{O}_{\mathcal{X}_s,x}$.

There exists an affine open subset $W \ni x$ and a regular element $g \in \mathcal{O}_{\mathcal{X}}(W)$ such that $g_x = f$, and that $V(g) \cap W_s = \{x\}$. Let Δ be the closure in $\mathcal{X}$ of an irreducible component of $V(g)$ passing through x . Then Δ is an irreducible horizontal divisor passing through x .

- (b) Let $E := V(g)$ and let W be as in (a). Then E can be seen as an effective Cartier divisor on W . Moreover, $E \rightarrow \text{Spec } \mathcal{O}_K$ is quasi-projective and quasi-finite because $\dim E_s < \dim \mathcal{X}_s = 1$. By a corollary of Zariski's Main theorem ([2, Corollary 4.4.8]), there exists an affine open neighborhood U of x such that $E \cap U$ is an open subscheme of a scheme Z that is finite over $\text{Spec } \mathcal{O}_K$.

$$E \cap U \xrightarrow{\text{open emb.}} Z \xrightarrow{\text{finite}} \text{Spec } \mathcal{O}_K$$

Let D be the connected component of Z containing x . Since $\mathcal{O}_K$ is a Henselian local ring, Z is a disjoint union of local schemes. Hence D is local, and consequently $D \subset E \cap U \subset \mathcal{X}$.

Finally, since D is closed in Z , it is finite over $\text{Spec } \mathcal{O}_K$. A finite morphism $D \rightarrow \text{Spec } \mathcal{O}_K$ between affine schemes is projective ([2, Exercise 3.3.22]), and therefore proper ([2, Theorem 3.3.30]). So the morphisms

$$D \rightarrow \mathcal{X} \rightarrow \text{Spec } \mathcal{O}_K \quad \text{and} \quad \mathcal{X} \rightarrow \mathcal{O}_K$$

are proper, which implies that $D \rightarrow \mathcal{X}$ is also proper ([2, Proposition 3.3.16(e)]). Consequently, D is closed in $\mathcal{X}$ and therefore a Cartier divisor, as required. $\square$

Now we can finally prove Proposition 2 (ii).

Proof (of Proposition 2 (ii)). Let $Z_1, \dots, Z_n$ be the irreducible components of $\mathcal{X}_s$ that induce the valuations in $V \subset V(\mathcal{X})$. By Lemma 12 (b), there exists an effective horizontal Cartier divisor D_i such that $\mathcal{X}_s \cap \text{Supp } D_i$ is a point of Z_i that belongs to none of the components of $\mathcal{E}$. Let $D = \sum_{i=1}^n D_i$. By virtue of Lemma 10, if necessary replacing D by a multiple, we can even suppose that $\mathcal{L} := \mathcal{O}_{\mathcal{X}}(D)$ is generated by its global sections.

The theorem then results from Proposition 9. Indeed, $\mathcal{L}|_E \simeq \mathcal{O}_E$ if $E \in \mathcal{E}$ since $E \cap \text{Supp } D = \emptyset$; if $E \notin \mathcal{E}$, then E is some Z_i and $\deg \mathcal{L}|_E \geq \deg \mathcal{O}_{\mathcal{X}}(D_i)|_{Z_i} > 0$, and hence $\mathcal{L}|_E \not\simeq \mathcal{O}_E$. $\square$

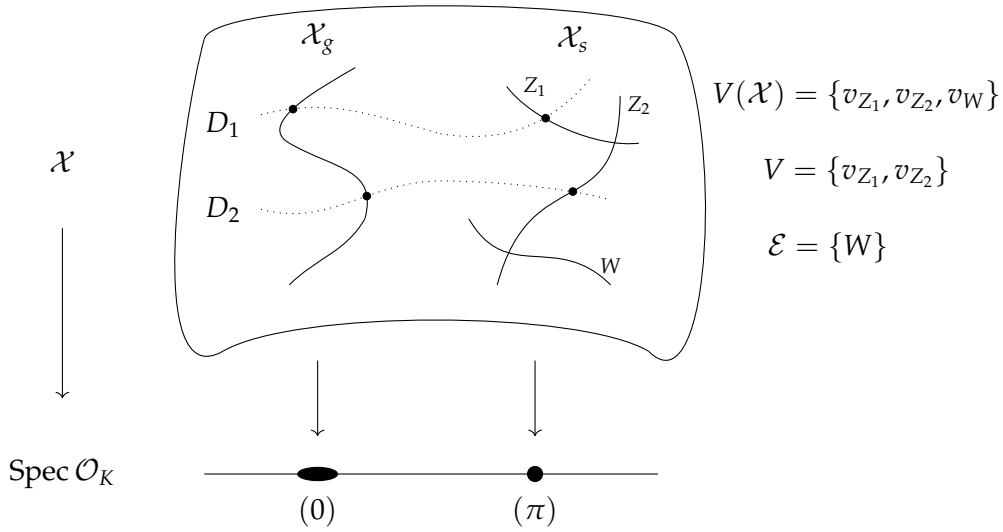


Figure 1 – Illustration of the proof of Proposition 2 (ii)

Remark 13 The proof of Proposition 2 (ii) shows that actually all models $\mathcal{X}$ of X are projective. Indeed one just has to choose $V = V(\mathcal{X})$ in the proof to obtain an effective Cartier divisor D which meets all of the irreducible components of the fiber $\mathcal{X}_s$. Since $\mathcal{X}_s$ is a projective curve ([2, Exercise 7.5.4]), $\mathcal{O}_{\mathcal{X}}(D)|_{\mathcal{X}_s}$ is ample ([2, Proposition 7.5.5]). It follows from [2, Corollary 5.3.24] that $\mathcal{O}_{\mathcal{X}}(D)$ is ample. So $\mathcal{X} \rightarrow \text{Spec } \mathcal{O}_K$ is projective ([2, Proposition 5.3.6]).

References

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